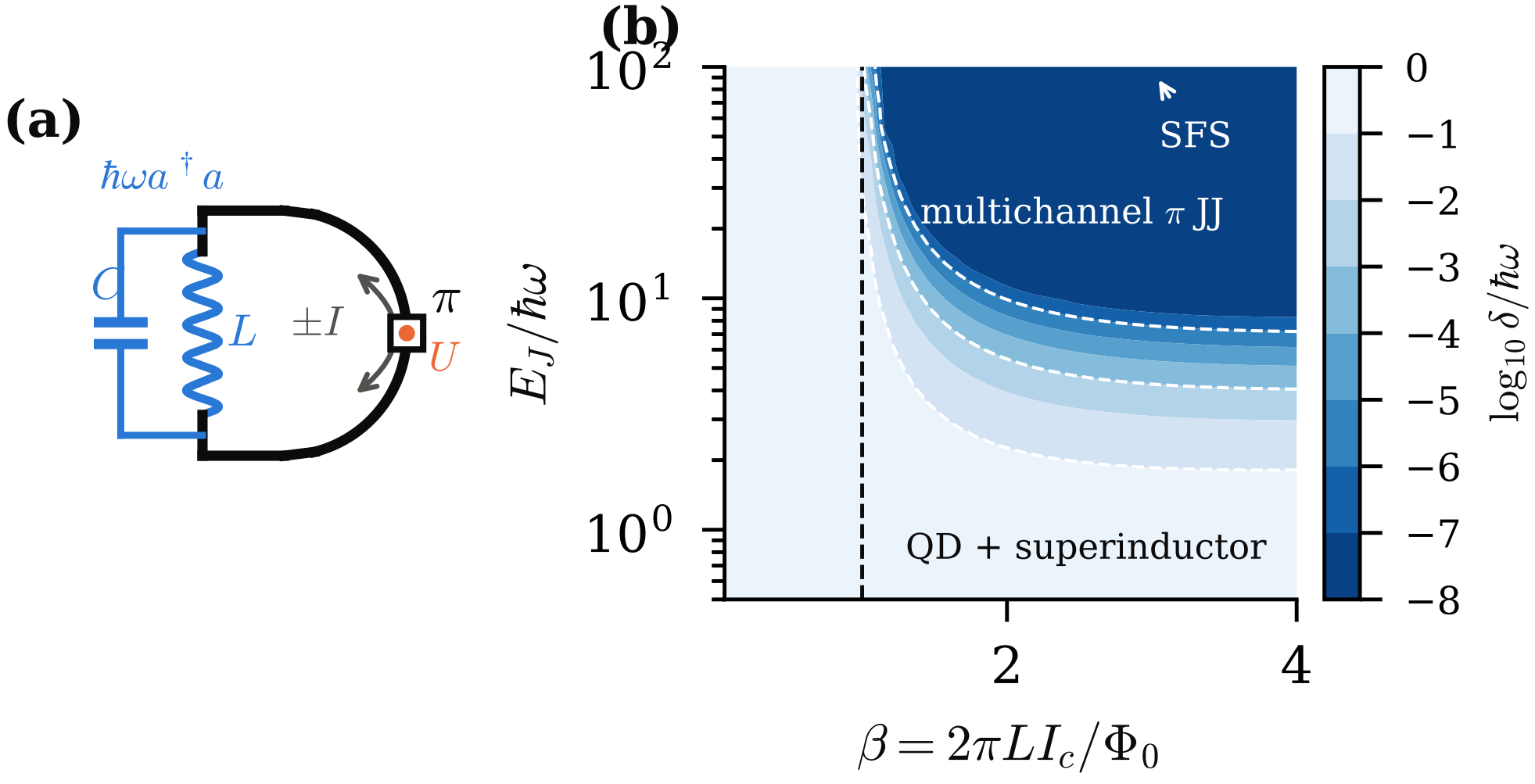
(a)
$\hbar\omega a^\dagger a$
C
L
$\pm I$
π
U
(b)
10^2
10^1
10^0
$E_J/\hbar\omega$
SFS
multichannel π JJ
QD + superinductor
2
4
$\beta = 2\pi L I_c/\Phi_0$
0
−1
−2
−3
−4
−5
−6
−7
−8
$\log_{10}\delta/\hbar\omega$